🌐 Module 13: IoT Penetration Testing

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# Exercise 03: Using Nmap DICOM Scripts
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Scenario

Healthcare IoT Auditing with Nmap Scripting Engine (NSE) utilizes specialized NSE scripts to discover, audit, and evaluate security posture across medical imaging devices running **DICOM** (Digital Imaging and Communications in Medicine) protocol services.

Key technical concepts in DICOM Nmap scanning:
1. **NSE Script Location & Inventory**: Locating built-in DICOM NSE scripts in **/usr/share/nmap/scripts/** using wildcard search (`ls *dicom*`).
2. **NSE Lua Logic Auditing**: Inspecting script internals (**dicom-ping.nse**) via text editors (`pluma`) to analyze PDU handshake verification and Application Entity Title (AET) probe parameters.
3. **Target Port Fingerprinting**: Scanning medical target servers (e.g., Orthanc DICOM server running on port **4242**) using **nmap -sV -p 4242 --script dicom-ping [Target_IP]**.
4. **Service Vulnerability Reporting**: Identifying open, unauthenticated DICOM daemon endpoints that expose sensitive Electronic Protected Health Information (ePHI) or lack encryption transport protocols.

In this exercise, we log into **Parrot Security** (**pentester** / **toor**), navigate to `/usr/share/nmap/scripts/` , list DICOM scripts, inspect `dicom-ping.nse` source code, execute `nmap` service detection and `dicom-ping` NSE audit against Ubuntu DICOM server on TCP port **4242**, and record service status reports.

Objective

The objective of this lab is to help students learn how to:
- Authenticate to Parrot Security target VM (**pentester** / **toor**)
- Navigate to Nmap scripts directory: **cd /usr/share/nmap/scripts**
- List available DICOM NSE scripts: **ls *dicom***
- Inspect NSE script logic using Pluma editor: **pluma dicom-ping.nse**
- Scan target Ubuntu medical server: **nmap -sV -p 4242 --script dicom-ping 192.168.0.x**
- Identify listening DICOM service port (**4242**) and assess security compliance

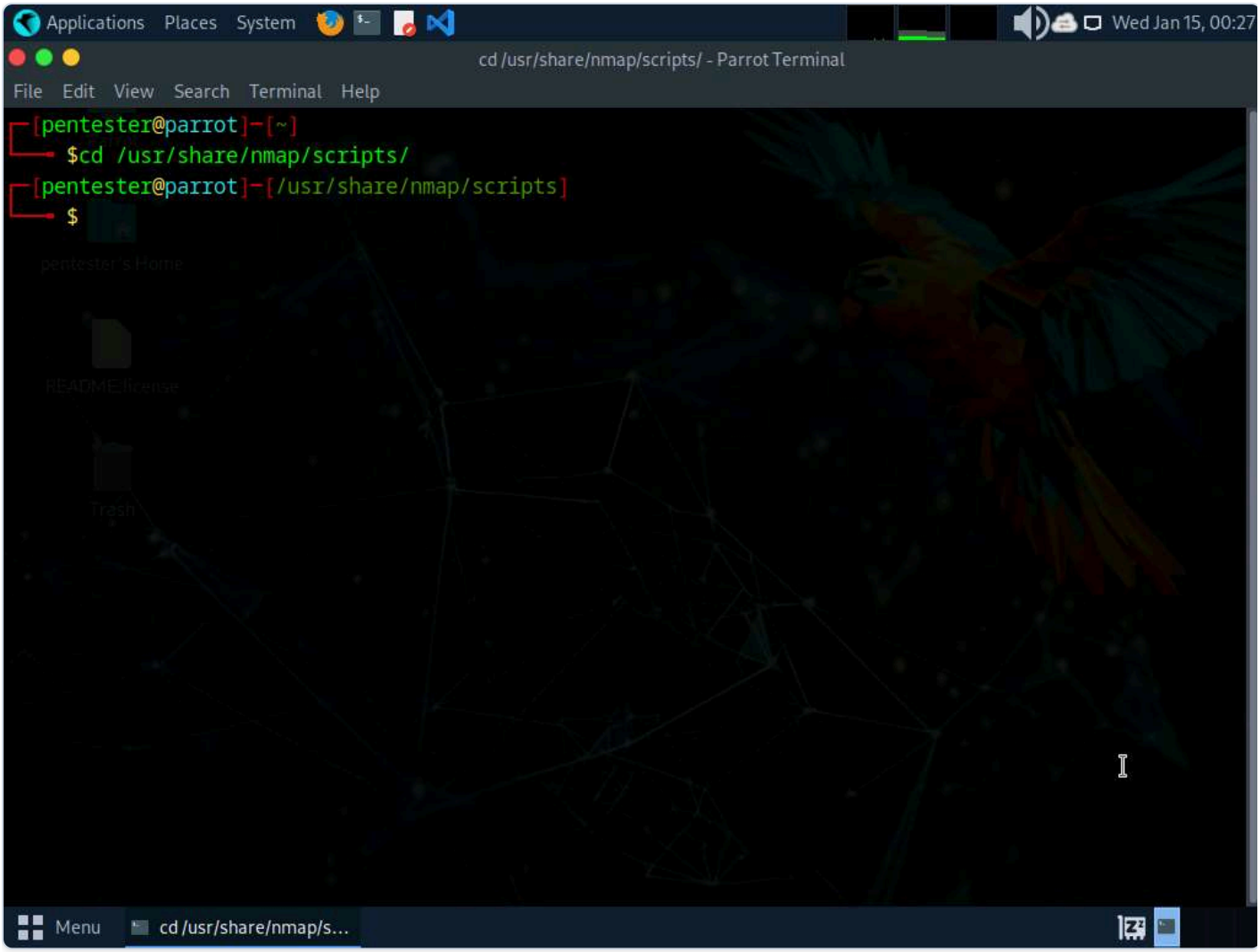
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Task 1: Inspect Nmap DICOM Scripts, Execute dicom-ping NSE Scan, and Audit Medical Server Port 4242

In this task, we list Nmap DICOM scripts in `/usr/share/nmap/scripts` , review `dicom-ping.nse` , and run service scanning against port `4242` .

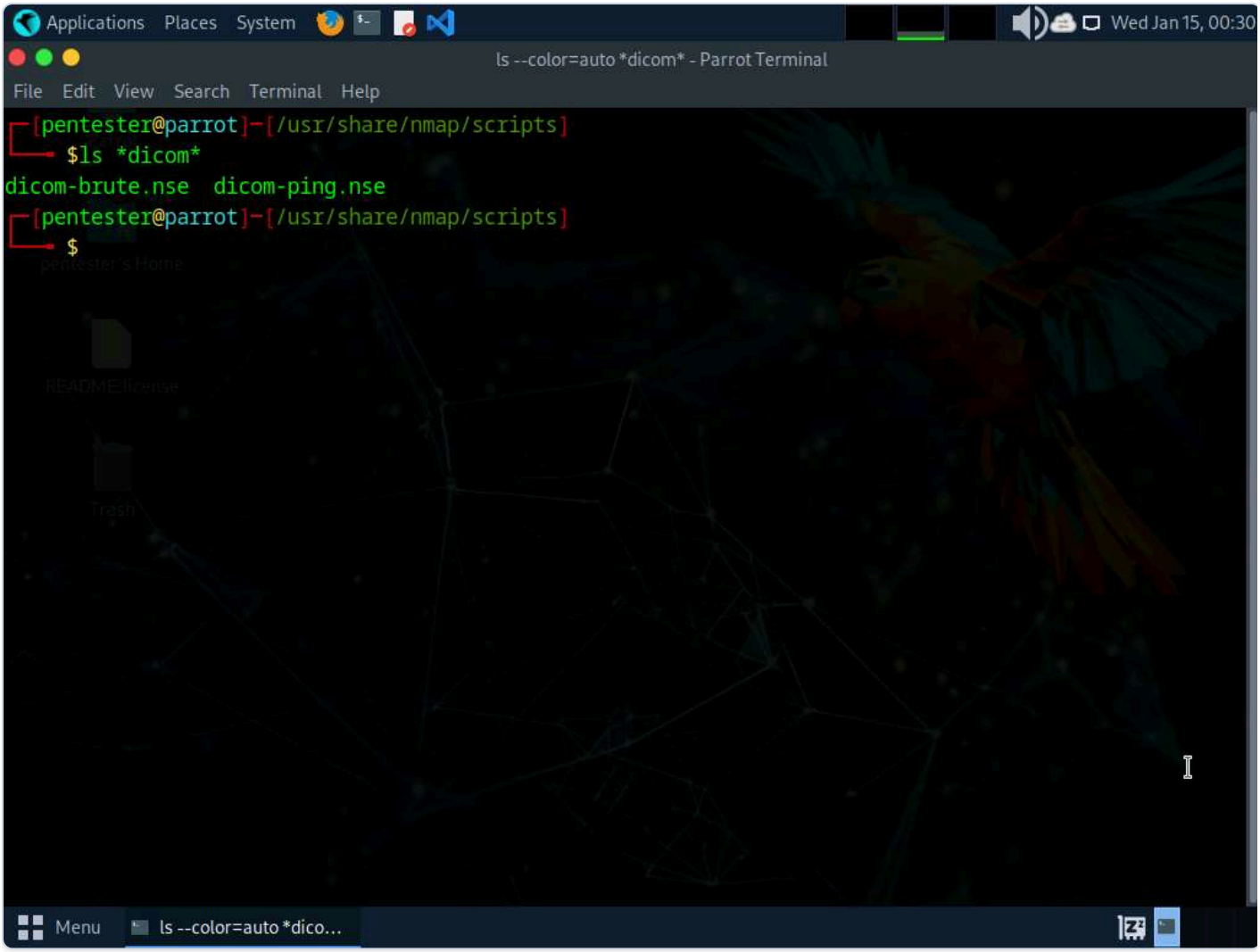
Lab Tasks

1. Log into **Parrot Security** VM using credentials (**pentester** / **toor**). Open terminal and navigate to Nmap script repository:
   cd /usr/share/nmap/scripts

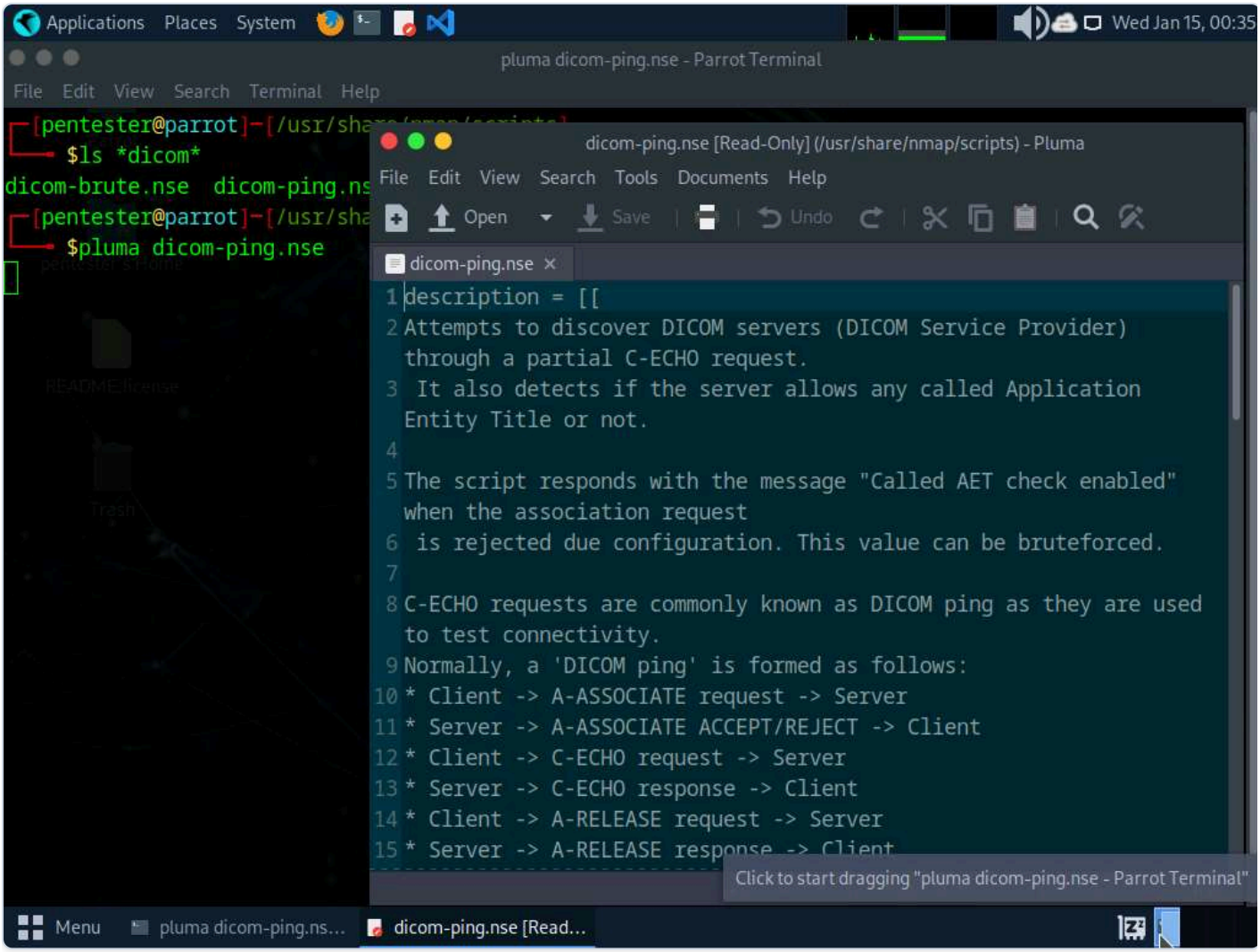


1. List all DICOM-related NSE scripts:

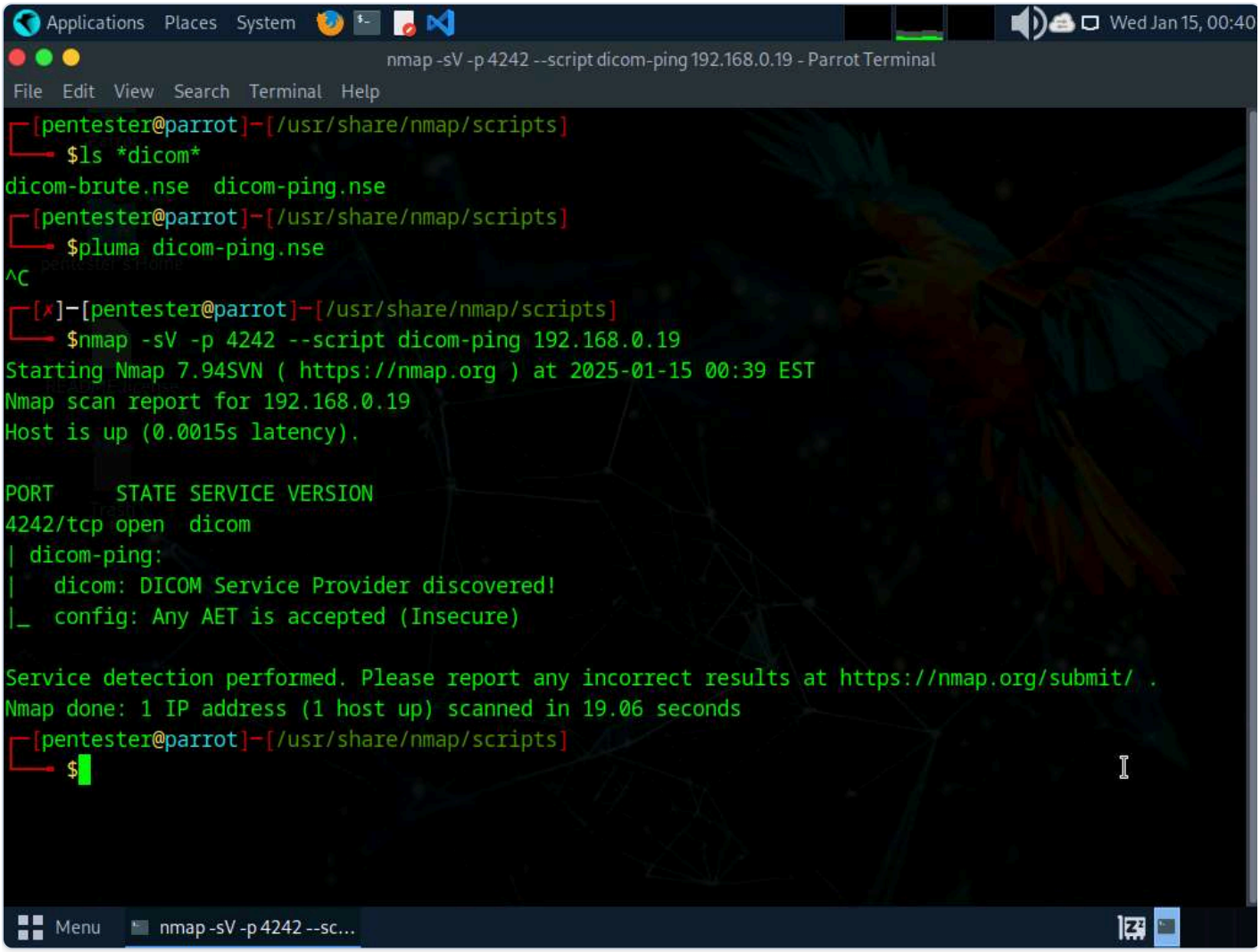
   ls *dicom*



1. Open **dicom-ping.nse** in Pluma text editor to inspect probe definitions:
   pluma dicom-ping.nse



1. Execute Nmap version scan with DICOM ping script against Ubuntu Orthanc server:
   nmap -sV -p 4242 --script dicom-ping 192.168.0.x
   Observe output: DICOM service detected on TCP port **4242**.



This concludes the lab exercise.

> **IMPORTANT:** ❓ **Question 13.3.1.1** On the Parrot Security machine, use nmap to detect DICOM server in ubuntu machine. Enter the port in ubuntu on which DICOM server is running. [!SUCCESS] ✅ **Answer:** **4242**